10.1.2 Language choice for test programs

Usually, testers or test programs are written in the same language as the main application to be tested. In this case, you can write the tester itself as a smart contract in the Solidity language. The Pet Shop example provided with the Truffle documentation illustrates tests written in Solidity (https://www.trufflesuite.com/tutorials/pet-shop). But many smart contracts, such as Ballot, use the `address` data type for the chairperson and the voters. This causes a problem when another smart contract is used as a tester. So we will use the alternative language supported by Truffle--JavaScript (JS)--for writing our tests. Truffle supports both languages and has tools to support JS-based testers, as well as JS test frameworks such as Mocha and Chai. These tools provide commands that are specially designed to write clean, expressive test code. You'll explore testing with the three Dapps developed in chapters 2-5. Choosing these familiar Dapps helps you focus on the testing aspects. In the following sections, you'll use

- The familiar smart contracts Counter.sol, Ballot.sol, and BlindAuction.sol
- Test commands (`it`, `describe`) of the Mocha test framework
- The Truffle assertion framework in Chai (`assert`)
- Test commands with reverting conditions within a smart contract